U = F(U, U)
The Recursive Binary Fold as the Theory of Everything

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1. Abstract

This paper proposes that the irreducible generative primitive of all structured reality is a recursive binary fold, expressed as the fixed-point equation $U = F(U, U)$. We argue that any universe containing at least one distinction—one boundary between *this* and *that*—necessarily instantiates a binary pairing operation, and we provide a formal argument that recursion is the only mechanism by which such an operation can generate unbounded complexity. For the fold to produce differentiation, directionality, and entropy, it must satisfy three algebraic constraints: non-commutativity, nonlinearity, and partial irreversibility. These three properties, combined with depth-bounded recursion and a termination condition, constitute the minimal viable architecture for a self-generating universe—a complete primitive we designate *Tuttle's triad*. The only logical structure that escapes binary recursion is the pure, structureless monad, which, having no internal distinctions, is operationally indistinguishable from nothing and therefore falls outside the domain of structured reality. This framework refines Wheeler's "it from bit" programme by specifying the exact generative mechanism that converts bit into it. We acknowledge that the framework is, in its present form, a meta-ontological foundation rather than a quantitative dynamical theory, and we trace its interpretive consequences across physics, biology, and cosmology without claiming specific numerical predictions.

2. Introduction: The Question of the Minimal Generative Primitive

The search for a "theory of everything" has historically been understood as a search for a unified Lagrangian—a single dynamical law from which the four fundamental forces and the particle spectrum can be derived. This paper takes a different, more foundational approach. Rather than

asking *what laws govern the universe*, we ask: *what is the simplest operation that, applied to itself, is sufficient to generate any structured reality whatsoever?*

We call this operation the *generative primitive*. A theory of everything, on this view, is not merely a theory of forces and particles but a theory of why there is structure at all—why there is *something* rather than undifferentiated nothing. The primitive must be logically minimal: it must not presuppose any structure it claims to generate.

The central claim of this paper is that the minimal generative primitive is a *recursive binary fold*: an operation that takes two inputs and produces one output, where both inputs are themselves products of the same operation. Expressed as a fixed-point equation: $U = F(U, U)$. The universe, on this account, is what you get when the fold operation is applied to itself, recursively, with the right algebraic properties.

A natural question arises at this stage: why must the primitive be recursive? Could a single distinction, combined with a static (non-recursive) rule, generate the complexity we observe? The answer is no, and the reason is fundamental. A static rule applied to a fixed input produces a fixed output; it may generate one level of structure but cannot generate a second level without re-applying itself to its own product. The moment it does so, it has become recursive. More precisely: any process that generates structures of unbounded depth must, by definition, apply its generative operation to its own prior outputs—and this is exactly what recursion is. A non-recursive primitive is bounded in the complexity it can produce; only recursion converts finite rules into potentially unbounded structural depth. The binary fold must therefore be recursive, not as an optional enhancement, but as a logical requirement for generativity beyond a single step.

This is a strong claim, and we wish to state its scope precisely. We do not claim to derive the Standard Model, general relativity, or any specific physical constants from this framework. We claim something both more modest and more fundamental: that recursive binary folding is the necessary and sufficient *form* of any generative process that produces structure from self-reference. Specific physical theories are particular instantiations of a fold satisfying particular boundary conditions. The fold itself is the invariant.

3. Logical Foundation: Why the Binary Recursive Fold Is Unavoidable

3.1 The Primacy of Distinction

Following Spencer-Brown (1969), we begin with the observation that the most primitive act of cognition, measurement, or existence is the drawing of a distinction. A distinction produces two sides: *this* and *not-this*. Before any physics, any mathematics, any ontology, there must be at least one boundary. The existence of a boundary is equivalent to the existence of a binary pair: two regions separated by a mark.

This is not a contingent fact about our universe; it is a logical necessity for any universe containing structure. Structure requires differentiation; differentiation requires boundary; boundary produces a pair. Therefore, the minimal alphabet of structured reality is binary.

We should acknowledge a subtlety here. Some philosophers have argued that a distinction can be non-generative—that drawing a boundary produces two static regions but does not, by itself, constitute an *operation*. This is correct, and it is precisely why distinction alone is insufficient. The distinction provides the alphabet (two sides); the fold provides the grammar (an operation on that alphabet). We must therefore make the logical chain explicit.

3.2 From Pair to Fold: A Formal Argument

The progression from distinction to recursive fold can be stated as a sequence of four steps, each of which is logically entailed by the previous:

Step 1 (Distinction → Pair). Any act of distinction produces exactly two regions. This is definitional: to distinguish is to create a binary partition.

Step 2 (Pair → Operation). For the two regions to participate in further structure—for the universe to contain more than one boundary—the regions must serve as inputs to a process that produces a new entity. That process is a binary operation: it takes two inputs (the regions) and yields one output (the composite). Without this step, the universe is frozen at exactly one distinction and two regions, with no mechanism for further differentiation.

Step 3 (Operation → Recursion). For the operation to generate structures of depth greater than one, its output must be eligible as input to the same operation. This is the definition of recursion. A non-recursive operation can produce exactly one level of composites; unbounded

structural depth requires self-application. As argued in Section 2, this is not optional but is the minimal condition for generativity beyond a single combinatorial step.

Step 4 (Recursion → Fixed Point). A recursive binary operation applied exhaustively to its own products defines a fixed-point equation: the totality U of all products satisfies $U = F(U, U)$, where F is the operation. U is the least fixed point of this equation—the smallest totality closed under F.

Each step is independently defensible, and the conjunction yields the central thesis: any structured reality necessarily instantiates a recursive binary fold whose totality is a fixed point.

3.3 The Monad Exception and Its Dissolution

There is exactly one logical structure that does not reduce to recursive binary folding: the pure monad—a structureless, undifferentiated unity with no internal distinctions. Because it contains no boundary, it contains no pair; because it contains no pair, no fold has occurred.

However, the monad's exemption is vacuous. A structureless monad has no properties, no internal states, no distinguishable features. It is, by every operational criterion, indistinguishable from nothing.[1] It cannot be observed, measured, or interacted with. It is not part of any *actual* reality precisely because "actual reality" is defined by the presence of structure, and structure requires distinction.

We must be precise about the epistemological status of this claim. We do not assert that the monad "does not exist" in some deep metaphysical sense; we assert that, under an operationalist criterion, it is indistinguishable from non-existence. A reviewer might object that "indistinguishable from nothing" is an epistemic claim (about what we can know) rather than an ontological one (about what is). We accept this distinction and note that our framework is deliberately operationalist: it characterises reality in terms of what can be structurally differentiated, not in terms of what might exist beyond all possible interaction. The monad may

[1]We adopt an operationalist criterion throughout: "exists" means "is in principle distinguishable from its negation by some operation." This is not a metaphysical commitment to anti-realism; it is a methodological choice that keeps the framework grounded in what can be structurally characterised rather than what might or might not "really" exist beyond all possible interaction.

"exist" in some metaphysical register inaccessible to structural analysis; if so, it exists outside the scope of any theory of structured reality, including this one.

The only way for the monad to enter reality is to break its own symmetry—to draw a distinction—at which point it instantiates the first binary fold and ceases to be a monad. The fold is therefore the gateway between non-existence and structured reality.

4. The Minimal Architecture: Tuttle's Triad

Having established that the recursive binary fold is logically unavoidable, we now specify the minimal architecture required for such a fold to generate a structured universe. This architecture consists of three components, which together constitute the complete generative primitive. We refer to this triad as *Tuttle's triad*, treating it not as a starting point for candidate fold architectures but as the complete and irreducible primitive itself.

4.1 Component 1: Self-Referencing Binary Recursion

The fold F must take exactly two inputs and produce exactly one output, and both inputs must be products of prior applications of F (or, at the base case, of the initial distinction). This is the engine of complexity: each level of recursion produces outputs that become inputs for the next level, generating a branching tree of increasingly composite structures.

The choice of binary—rather than ternary or n-ary—pairing is not arbitrary. As established in Section 3, the minimal act of distinction is binary. Any n-ary operation with $n > 2$ can be decomposed into a sequence of binary operations (just as any n-ary tree can be encoded as a binary tree via standard Schönfinkelisation). The binary fold is therefore the irreducible case; higher-arity operations are composites.

4.2 Component 2: Depth-Bounded Termination Condition

Unbounded recursion without a termination condition produces no stable structure—it is the computational analogue of a divergent series. For the fold to generate a universe with identifiable, persistent features, there must be a mechanism that halts or stabilises recursion at certain depths.

The termination condition need not be externally imposed. It can arise from the fold's own dynamics: a fold that produces diminishing returns—outputs increasingly similar to inputs—will

naturally approach fixed points at finite depth. Physically, this corresponds to the existence of stable structures—particles, atoms, organisms—that persist without further decomposition at their characteristic scale.

To make this concrete, consider a minimal illustrative example. Define a fold $F(a, b) = (a + b) / 2 + \varepsilon(a - b)$ for some small asymmetry parameter ε. When $a \approx b$, the fold's output is approximately equal to its inputs: the difference term vanishes, the averaging term stabilises, and recursion halts at a fixed point. This is a toy model, not a physical proposal; its purpose is to demonstrate that self-termination is an intrinsic possibility of asymmetric folds, not an additional axiom. Real physical termination (e.g., at the Planck scale) would emerge from the specific algebraic form of the actual fold, not from an externally imposed cutoff.

The Planck scale itself may be understood, within this framework, as the deepest recursion depth at which the fold produces operationally distinguishable outputs. Below this depth, the fold's products are indistinguishable—not because space "ends," but because the fold's resolution saturates.

4.3 Component 3: Asymmetric Fold

The fold $F(a, b)$ must be asymmetric: the two inputs must not be interchangeable without altering the output. This is the most physically consequential requirement, and we develop its algebraic implications in the next section. Without asymmetry, the fold produces only symmetric, undirected, entropy-free structures—which is to say, no physics at all.

Together, these three components—recursion, termination, and asymmetry—form the complete primitive. No one component is sufficient alone; all three are necessary; and their conjunction is sufficient to generate structured reality. This is the claim of Tuttle's triad: the irreducible, self-contained generative kernel from which any structured universe must arise.

5. Algebraic Requirements on the Fold

We now examine the specific algebraic properties that the fold F must satisfy. These are not additional postulates; they are consequences of the requirement that the fold generate a universe with directionality, differentiation, and irreversible history.

5.1 Non-Commutativity

A commutative fold satisfies $F(a, b) = F(b, a)$ for all inputs. Such a fold cannot distinguish left from right, before from after, cause from effect. Since our universe manifestly exhibits all of these directional asymmetries, the generative fold must be non-commutative: $F(a, b) \neq F(b, a)$ in general.

Non-commutativity is the algebraic expression of the physical fact that order matters. It is the origin of chirality, of temporal asymmetry, and of the causal structure of spacetime. In the language of abstract algebra, the fold generates a non-abelian structure—a feature shared by the gauge groups of the Standard Model (SU(3) and SU(2) are both non-abelian). We note this structural resonance without claiming a derivation: the framework predicts that the deepest physical symmetries must be non-abelian *because* the generative fold is non-commutative, but deriving the specific gauge groups from a particular fold would require identifying the fold's exact algebraic form.

5.2 Nonlinearity

A linear fold satisfies $F(\alpha a + \beta b, c) = \alpha F(a, c) + \beta F(b, c)$. Linear operations preserve superposition; they cannot create genuinely new structure from existing components. More precisely, a linear operation cannot generate new information: its output is fully determined by, and decomposable into, a weighted sum of its inputs. Novelty—the emergence of composite entities whose properties are not mere sums of their parts—requires that the fold be nonlinear.

In physical terms, nonlinearity is responsible for bound states, phase transitions, spontaneous symmetry breaking, and the rich phenomenology of interacting systems. A linear fold would produce a universe of non-interacting waves—a free field theory with no particles, no atoms, and no chemistry. The requirement of nonlinearity is therefore not a mathematical technicality but the minimal condition for a universe containing anything other than superposed plane waves.

5.3 Partial Irreversibility

A fully reversible fold is one for which F admits a global inverse: given the output, both inputs can always be recovered. Such a fold conserves all information and produces no entropy—no arrow of time, no thermodynamics, no aging, no death.

Our universe is not like this. The second law of thermodynamics is, by all available evidence, universally valid. Therefore the fold must be at least *partially* irreversible: some applications of F produce outputs from which the exact inputs cannot be fully recovered. Information is not destroyed but compressed, coarse-grained, or rendered practically inaccessible—consistent with both classical thermodynamics and the holographic bound on information density.

The qualification "partially" is important. A *fully* irreversible fold—one that destroys all information about its inputs—would produce a universe with no persistent structure, no memory, no identity over time. The fold must occupy the regime between full reversibility and full irreversibility: it must lose enough information to generate an arrow of time but retain enough to support stable structure and physical law.

We may summarise the algebraic constraints as a conjunction: the fold must be non-commutative *and* nonlinear *and* partially irreversible. Removing any one of these constraints collapses the generative capacity of the fold. Non-commutativity without nonlinearity produces directed but interaction-free systems. Nonlinearity without non-commutativity produces interacting but undirected systems. Either without partial irreversibility produces a timeless, entropy-free block universe. All three together are necessary and jointly sufficient.

6. Relation to Prior Work

6.1 Wheeler's "It from Bit"

Wheeler (1990) proposed that the physical world derives its existence from information-theoretic origins: every "it"—every particle, every field, every spacetime event—derives its existence from binary yes-or-no answers to observational questions. This programme correctly identifies the binary character of foundational reality but leaves the generative mechanism unspecified. Wheeler tells us that reality is *made of* bits; he does not tell us how bits *make* things.

The recursive binary fold provides the missing mechanism. It is the operation by which bits compose into structures: two bits fold into a composite, which itself becomes a bit available for further folding. "It from bit" becomes "it from *fold of* bit." The fold is the verb that Wheeler's noun requires.

6.2 Tuttle's Fold Architectures

The present work extends Tuttle's earlier investigations into binary fold architectures, which explored the space of possible fold operations satisfying various algebraic constraints.[2] Those investigations catalogued candidate folds and analysed their generative capacities but treated the triad's three components as independently variable design parameters—allowing, for instance, the study of symmetric folds or fully reversible folds in isolation. The present paper crystallises the conclusion of that programme: the three components are not independently variable but jointly necessary and jointly sufficient. Their conjunction is not one promising candidate architecture among many; it is the unique irreducible primitive. The earlier work mapped the design space; this paper identifies its unique minimum.

6.3 The Holographic Principle

The holographic principle, arising from black hole thermodynamics and formalised by 't Hooft (1993) and Susskind (1995), states that the information content of a spatial region scales with its boundary area rather than its volume. In the fold framework, this is a natural consequence of recursive depth-bounding: the "surface" of a recursion tree (its leaves, where termination conditions are met) encodes the information about the interior.

To give this connection more formal substance: a binary recursion tree of depth d has at most 2^d leaves but contains $2^{d+1} - 1$ total nodes. The ratio of surface (leaf) information to volume (total node) information approaches 1/2 as d grows, and the absolute leaf count grows exponentially with depth while being strictly determined by a two-dimensional parameter (depth itself). The holographic bound, on this view, is an expression of the fold's finite recursion depth—a feature of its termination condition, not an additional physical law.

6.4 p-Adic and Ultrametric Geometry

The recursive binary fold naturally generates ultrametric (tree-like) distance structures rather than Archimedean (flat) ones. In a recursion tree, the "distance" between two leaves is determined by

[2]The earlier work catalogued a space of candidate fold operations and analysed their generative capacities under varying algebraic constraints, but treated the triad's three components as independently variable parameters. The present paper's contribution is the argument that the three components are not independently variable but jointly necessary and jointly sufficient—that removing any one collapses the generative capacity entirely, and that their conjunction is the unique minimal architecture.

the depth of their most recent common fold ancestor—a structure that is formally p-adic. This is significant because p-adic and ultrametric methods have appeared independently in several areas of theoretical physics, including spin glass theory, the AdS/CFT correspondence, and certain approaches to quantum gravity. The fold framework suggests that these appearances are not isolated mathematical coincidences but reflections of the recursive architecture underlying physical structure. A diagrammatic representation of this tree-to-ultrametric correspondence—showing how fold depth maps to p-adic valuation—would be a natural addition to a future extended treatment.

6.5 Lambda Calculus and Computational Foundations

The fixed-point equation $U = F(U, U)$ is closely related to the fixed-point combinators of the untyped lambda calculus, particularly the Y combinator.[3] This is not an analogy; it is a structural identity. The fold is a computational operation, and the universe it generates is, in a precise sense, a computation. This does not commit us to a "simulation hypothesis"—it commits us only to the claim that the structure of reality is isomorphic to the structure of a self-applying recursive function. Whether this computation runs on any external substrate is a question the framework does not address and does not need to address.

6.6 Category Theory and Coalgebra

In category-theoretic terms, the recursive binary fold defines a coalgebra: a structure that is characterised by its unfolding behaviour rather than its constructive composition. The fixed-point $U = F(U, U)$ is the terminal coalgebra of the binary functor F. This connection is mathematically precise and provides a rigorous foundation for the framework within existing formal systems. The apparatus of coalgebraic recursion—bisimulation, coinduction, final semantics—applies directly to the fold and offers tools for analysing its properties without the need for new mathematical foundations.

7. Implications for Physics, Biology, and Cosmology

[3]Crucially, fixed-point combinators such as the Y combinator are intrinsic to the calculus itself and do not require an external evaluator or runtime environment. The fixed point is a property of the algebraic structure, not of any machine that computes it. This is significant because it means U = F(U, U) can be self-sustaining: no external substrate is required to "run" the recursion.

We trace several interpretive consequences of the fold framework, emphasising structural correspondences rather than specific numerical predictions. The framework operates at the level of meta-ontology—it specifies the *form* of physical law, not its particular parameters. The implications discussed below are interpretive consequences of the architecture, not empirical predictions in the falsificationist sense. Each would require substantial further development before it could be tested against alternatives.

7.1 Physics

If the fold is the generative primitive, then the gauge symmetries of the Standard Model are not fundamental but emergent: they are the residual symmetries of a particular fold architecture at a particular recursion depth. The non-abelian character of SU(3) and SU(2) reflects the non-commutativity of the underlying fold; the abelian U(1) may represent a limiting case where the fold locally symmetrises. The hierarchy problem—the vast disparity between the electroweak scale and the Planck scale—may admit interpretation as a consequence of the exponential growth of structure across recursion levels, where each fold level introduces a multiplicative factor.

Quantum mechanics, in this framework, admits a natural interpretation as the description of systems at recursion depths where the fold has not yet terminated—where the binary pairing has not yet resolved into a definite classical outcome. The wavefunction, on this reading, is not a physical field but a description of an in-progress fold: a superposition of potential pairings that has not yet committed to a specific asymmetric configuration. Measurement corresponds to fold-termination. The Born rule may emerge from the statistics of fold-resolution in a non-commutative, partially irreversible algebra—though we emphasise that this is a suggestive structural correspondence, not a derivation, and that rigorously obtaining the Born rule from fold dynamics remains an open problem.

7.2 Biology

Biological systems exhibit the fold architecture with striking transparency. DNA replication is a binary fork: one double helix unfolds into two. Cell division (mitosis, meiosis) is recursive binary folding. Phylogenetic trees are recursion trees. Protein folding—despite its name—is the physical process of a linear chain exploring a tree of conformational pairings until a termination condition (thermodynamic stability) is reached.

The ubiquity of binary branching in biological architecture is, on the present account, not a contingent feature of terrestrial evolution but a necessary consequence of the fold's status as the generative primitive. Any self-replicating, information-bearing system must instantiate binary recursion because that is the only mechanism by which structured complexity self-generates.

7.3 Cosmology

The cosmological Big Bang admits an interpretive correspondence within the fold framework: it may be understood as the initial fold—the first breaking of the monad's symmetry, the first distinction drawn within undifferentiated unity. We stress that this is a structural interpretation, not a predictive model. We do not claim to derive the Big Bang from fold dynamics; we observe that the fold framework provides a natural ontological vocabulary for describing it.

Inflation, on this reading, corresponds to rapid recursion—the fold applying to itself at an accelerating rate before termination conditions (stable particle physics) begin to bind. The large-scale structure of the universe—cosmic filaments, voids, clusters—exhibits the branching, hierarchical topology characteristic of a recursion tree.

The arrow of time is a direct consequence of the fold's partial irreversibility. Time does not "flow"; rather, the fold's asymmetric, partially irreversible character generates an ordering on fold-events that we experience as temporal succession. Entropy increase is the accumulation of irreversible folds; the heat death of the universe is the state in which no further asymmetric folding is possible—the return to a monad-like state, approached asymptotically but, if the fold is only partially irreversible, never fully reached.

8. Discussion: Scope, Apparent Exceptions, and Limitations

8.1 Apparent Non-Binary and Symmetric Systems

A natural objection is that many physical systems appear to be non-binary, non-hierarchical, or symmetric. Ternary and higher-order interactions, continuous symmetries, and flat (non-hierarchical) structures are ubiquitous. We address this class of objection directly.

Apparent non-binary systems are, almost without exception, *derivative*: they arise from one or more prior binary folds that then symmetrised, combined, or evolved. A ternary interaction,

for example, decomposes into two sequential binary pairings. A continuous symmetry group is the closure of a countable sequence of discrete (binary) operations. A flat network is a tree in which the branching has been obscured by the addition of cross-links—links that are themselves products of subsequent folds connecting previously separate branches.

The key insight is that *apparent* symmetry does not imply *fundamental* symmetry. The Standard Model's gauge symmetries are explicitly broken (electroweak symmetry breaking) or confined (colour confinement). The smooth, continuous manifold of general relativity is expected, on most approaches to quantum gravity, to be an approximation to a discrete substructure. In each case, the symmetric surface dissolves into asymmetric binary operations at sufficient depth.

8.2 What the Framework Does Not Do

Intellectual honesty requires a clear statement of what this framework does *not* accomplish.

First, it does not derive specific physical constants. The fine-structure constant, the cosmological constant, and particle masses do not fall out of the fold equation. These would require specifying the *particular* fold—the specific form of F—operative in our universe, which is beyond the scope of this paper.

Second, it does not yet provide a dynamical theory. The framework specifies the *architecture* of reality but not its dynamics—not the rate or schedule of folding, not the mechanism by which specific fold-resolutions are selected. A dynamical theory of folding would be a necessary next step toward contact with experimental physics.

Third, the framework is, in its current form, a meta-ontological claim rather than a falsifiable physical theory. It asserts that the *form* of all physical law is recursive binary folding, but it does not generate specific predictions that could be tested against alternatives. This is a genuine limitation, and we do not wish to disguise it. The framework's value at this stage is conceptual and organisational: it provides a unified vocabulary for phenomena that are currently described in separate formal languages. A falsifiability pathway—most likely through the construction of toy dynamical models whose predictions differ from those of standard frameworks—is a priority for future work but is not yet available.

8.3 Reflexivity and Self-Application

An important meta-theoretic observation: the fold framework applies to itself. This paper is a binary fold—a pairing of prior ideas into a new composite. The act of reading it is a fold: the reader's prior understanding is paired with the paper's claims to produce a new understanding. The scientific enterprise is a recursion tree of folds. This reflexivity is not a defect or a paradox; it is a consistency check. A theory of everything that did not apply to the process of theorising would be self-undermining.

9. Conclusion

We have argued that the recursive binary fold, expressed as the fixed-point equation $U = F(U, U)$, is the irreducible generative primitive of all structured reality. The argument proceeds from the logical necessity of distinction (Section 3), through the minimal architecture required for structure-generation (Section 4), to the algebraic constraints that the fold must satisfy (Section 5). We have formalised the logical chain from distinction to fixed point as a four-step argument and provided an illustrative example of self-terminating fold dynamics.

The framework's strength is its parsimony. It posits one operation (binary pairing), one structural requirement (recursion with termination), and one algebraic constraint set (non-commutativity, nonlinearity, partial irreversibility). From these, it claims to encompass the form of all structured reality—physical, biological, computational, and mathematical.

Its weakness, candidly acknowledged, is its current distance from quantitative physics. The gap between "the form of all physical law is recursive binary folding" and "the electron's mass is 0.511 MeV" is vast, and we have not bridged it. Bridging it would require identifying the specific fold operative in our universe—a programme that may require decades of further work, beginning with the construction of toy dynamical models and the mapping of fold-depth to physical scales.

Nevertheless, we maintain that the fold is the correct foundation. It sits beneath Wheeler's "it from bit" by specifying the exact mechanism by which bits compose into its. It unifies the binary recursion visible in computation, biology, and cosmology under a single generative operation. And it does so with the minimal logical apparatus: one operation, applied to itself, asymmetrically, with a stopping condition.

The theory of everything is not a Lagrangian. It is a fold.

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